

1.4 Vector Field

We have defined a vector field as a vector function of a position vector. In general, the magnitude and direction of the function will change as we move throughout the region, and the value of the vector function must be determined using the coordinate values of the point in question. Because we have considered only the rectangular coordinate system, we expect the vector to be a function of the variables x , y , and z .

If we again represent the position vector as $\mathbf{r}$, then a vector field $\mathbf{G}$ can be expressed in functional notation as $\mathbf{G}(\mathbf{r})$; a scalar field T is written as $T(\mathbf{r})$.

Example: A vector field $\mathbf{S}$ is expressed in rectangular coordinates as

$$\mathbf{S} = \frac{125}{[(x-1)^2 + (y-2)^2 + (z+1)^2]} \{ (x-1)\mathbf{a}_x + (y-2)\mathbf{a}_y + (z+1)\mathbf{a}_z \}$$

- (a) Evaluate $\mathbf{S}$ at $P(2, 4, 3)$.
- (b) Determine a unit vector that gives the direction of $\mathbf{S}$ at P .
- (c) Specify the surface $f(x, y, z)$ on which $|\mathbf{S}| = 1$

Solution: a) $5.95\mathbf{a}_x + 11.90\mathbf{a}_y + 23.8\mathbf{a}_z$
b) $0.218\mathbf{a}_x + 0.436\mathbf{a}_y + 0.873\mathbf{a}_z$
c) $\sqrt{(x-1)^2 + (y-2)^2 + (z+1)^2} = 125$